

Assignment #2

Assignment #2 Due Date: Feb 5, 2026 (by 23: 59 pm)

PART I (60 points)

Introduction

SHERPA Underground (Costmine) is a comprehensive application that merges the dataset of Mining Cost Service with industry-standard engineering practices. Costmine's practice has been well-recognized in the mining industry.

Website: <https://underground.costmine.com/>

Objective: The learning objective of this tutorial is to familiarize students with the layout of SHERPA Underground (Costmine) software and use the software for cost estimation.

Problem Description:

Please estimate the costs of mining a vein deposit that averages approximately 4,000 feet long, 2,000 feet high, and 12 feet thick. The deposit dips at an angle of about 75 degrees and contains approximately 8,200,000 tons of resources. It is located 4,500 feet horizontally from the closest point of access and lies 800 vertical feet below the surface. Initially, the deposit will be accessed by an **adit**. It will take about 2 years ("Year-1 and Year-2") to drive this adit, develop enough stopes to allow initial production, and construct all surface facilities. For the first 3 years ("Year-3, Year-4, and Year-5") of operation, ore will be mined at a rate of 800 tons per day. The mine will operate on a schedule of 10 hours per shift, 2 shifts per day, 320 days per year. An estimated 80 percent of production will come from Cut & Fill stopes with the remaining 20 percent recovered using Shrinkage stoping methods. Initially, ore and waste will be transported to the surface using Rear-Dump Trucks. During the second and third year of production ("Year-4 and Year-5"), a **second adit** will be constructed to allow an increase in production. For the 8 years following completion of this adit, the production rate will increase to 1,600 tons per day. Production from Cut & Fill stopes will decrease to about 60 percent, while production from the Shrinkage stopes will increase to 40 percent. During the final two years ("Year-12 and Year-13") of this

production segment (Segment #2), a **shaft** will be sunk to recover ore from lower levels. Upon completion of the shaft, ore production will decrease to 1,200 tons per day, and 100 percent of production will come from Cut & Fill stopes.

Example -cut-and-fill
Date: July 24, 2024 Created By: Metrics: Imperial Data Year: 2023

ADD SEQUENCE SAVE

Sequence 1  

Resource Mined This Sequence: 768,000 Tons

Project
Total Mineable Resources
8,200,000  Tons

Hrs/Shift  10.0
Shifts/Day  2
Days/Year  320

Daily Production Rate  800 Tons/Day
Haul To Surface
Rear-Dump Truck  Re...

From The Beginning Of Year  3
To The End Of Year  5

Primary Stoping
Method
Cut & Fill  Cut & Fill

Muck Collection  Slusher

Haul To Transfer Point
Rear-Dump Truck  Rear-Dump Truck

Production
80.0  Percent

Secondary Stoping
Method
Shrinkage  Shrinkage

Muck Collection  Scoop Tram

Haul To Transfer Point
Scoop Tram  Scoop Tram

Production
20.0  Percent

Example -cut-and-fill

Date: July 24, 2024 Created By: Metrics: Imperial Data Year: 2023



ADD SEQUENCE

SAVE

SEQUENCE 1

SEQUENCE 2

SEQUENCE 3

Sequence 2



Resource Mined This Sequence: 4,096,000 Tons

Project

Remaining Mineable Resources
7,432,000 Tons

Hrs/Shift

10.0

Shifts/Day

2

Days/Year

320

Daily Production Rate

1,600 Tons/Day

Haul To Surface

Rear-Dump Truck

Rear...

From The Beginning Of Year

6

To The End Of Year

13

Primary Stopping

Method

Cut & Fill

Cut & Fill

Muck Collection

Slusher

Slusher

Haul To Transfer Point

Rear-Dump Truck

Rear-Dump Truck

Production

60.0 Percent

Secondary Stopping

Method

Shrinkage

Shrinkage

Muck Collection

Scoop Tram

Scoop Tram

Haul To Transfer Point

Scoop Tram

Scoop Tram

Production

40.0 Percent

DELETE SEQUENCE

Example -cut-and-fill

Date: July 24, 2024 Created By: Metrics: Imperial Data Year: 2023



ADD SEQUENCE

SAVE

SEQUENCE 1

SEQUENCE 2

SEQUENCE 3

Sequence 3



Resource Mined This Sequence: 3,456,000 Tons

Project

Remaining Mineable Resources
3,336,000 Tons

Hrs/Shift

10.0

Shifts/Day

2

Days/Year

320

Daily Production Rate

1,200 Tons/Day

Haul To Surface

Hoist

Hoist

From The Beginning Of Year

14

To The End Of Year

22

Primary Stopping

Method

Cut & Fill

Cut & Fill

Muck Collection

Slusher

Slusher

Haul To Transfer Point

Rear-Dump Truck

Rear-Dump Truck

Production

100.0 Percent

Secondary Stopping

Method

Name

Name

Muck Collection

Name

Name

Haul To Transfer Point

Name

Name

Production

0 Percent

DELETE SEQUENCE

Example -cut-and-fill
 Date: July 24, 2024 Created By: Metrics: Imperial Data Year: 2023

ADD SEQUENCE SAVE

SEQUENCE 1 SEQUENCE 2 SEQUENCE 3

Sequence 1

Deposit

Geometry: Vein Average Dip: 75.0 Degrees

Average Length (Maximum Horizontal): 4,000.0 Feet Average Width (Minimum Horizontal): 12.0 Feet

Average Height (Vertical): 2,000.0 Feet Vertical Distance to Surface: 800 Feet Horizontal Distance to Surface: 4,500 Feet

DELETE SEQUENCE

Notes from the software

- At this point, the program suggests values for all remaining entries. We have used standard engineering techniques and data from a statistically significant number of operating mines to provide these values. However, since so much of underground mine design is determined by unique conditions specific to individual sites and since segments of each design can be somewhat subjective in nature, we encourage users to work carefully through all the screens to ensure that the model constructed by the program is representative. The program does not optimize any underground function, but simply determines what is needed and provides reasonable suggestions.
- Note the Density, Swell, Rock Quality Designation, and Compressive Strength values suggested by the program. These values are based upon the mining methods entered by the user and are derived statistically.
- We urge the user, as time permits, to carefully examine all of the remaining Input Data, Equipment, and Engineering screens to note the available input options.
- Final design is left to you. It is our hope, however, that we have provided a format in which a myriad of design options can be examined in a reasonable amount of time, allowing you to arrive at a practical final design option.

Report questions

- Please do a hand calculation to validate the numbers of “Remaining Movable Resources” in Sequences #2 and 3.
- In the software chart, there is a relationship between “Resource size, tons” and “Production, tpd.” Please use hand calculations to verify the relationship. Is this relationship accurate?

Enter the total amount of ore excavated and transported daily. Guideline production rates calculated using Taylor's Rule (an empirical formula sometimes used to relate min life to resource size) are:

Resource size, tons.....	Production, tpd
100,000.....	Shifts 80 y
1,000,000.....	450
5,000,000.....	1,500
10,000,000.....	2,500
50,000,000.....	8,400
100,000,000.....	14,000
500,000,000.....	46,000

- Using Taylor's Rule

Taylor's Rule: Life of Mine = $0.2 \times (\text{expected ore tonnes})^{0.25} \pm 1.2$ years

to hand calculate the “Life of Mine” for Sequences #1, 2, and 3. Do these calculated numbers match the defined “Life of Mine” in this project?

- As a design engineer, how do you decide the “Life of Mine?” Please have a discussion.
- Add Adit #2 using a similar dimension (compared with Adit #1) and 500 ft rock bolts at the “Portal Elevation” of 1300 ft.

- Add one shaft (per “Problem Description”) using the following settings:
 - Circular
 - Footwall
 - Production shaft
 - 1 compartment
 - Face area: 1500 ft²
 - Depth: Refer to deposit dimension
 - Concrete Reinforcement along the whole shaft.
- Report the “Capital Costs” to build Adit #2 and the Shaft.
- What are the **Total Capital Cost and Total Operating Cost** for the whole project?
- Copy the “Executive Summary” of three sequences to the report.
- There is no need to worry about the format of this assignment. Free style.
- Please submit the pdf copy to the Canvas Assignment before the due time.

PART II: Selection of Underground Mining Methods (40 points)

The conditions of the hangwall, orebody, and footwall are given by a hard rock mine.

- Shape: Tabular
- Average Thickness: 88 m
- Dipping angle: 85°
- Grade Distribution: Erratic
- Average deposit depth (to ground surface): 1200 m
- Average specific gravity of overburden: 2.7

- Average uniaxial compressive strength (UCS):
 - Hanging wall: 91 MPa
 - Ore: 103 MPa
 - Footwall: 125 MPa
- Average Rock Quality Designation (sampled by mine geologists):
 - Hanging wall: 27%
 - Ore: 50%
 - Footwall: 72%
- Fracture strength (sampled by mine geologists): Moderate for both host rock and ore

As a mine engineer,

- (1) Please hand calculate the rock substance strength of the hanging wall, ore, and footwall. Write down the process in this assignment. (20 points)
- (2) Please decide the best two underground mining methods using the UofA Macdonald Mining Method Selection Tool (<https://sites.ualberta.ca/%7Ewvliu/Mining%20Method%20Selection.html>). (20 points)
 - a. What are the scores of the two underground methods?
 - b. Please make a snapshot of the whole screen of the tool.